

MAKHACHKALA, Russian Federation

WMO: 374720

Lat: 43.000N

Lon: 47.500E

Elev: -19

StdP: 101.55

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.1	-8.6	-15.1	1.0	-11.8	-11.5	1.4	-7.8	13.5	2.1	11.4	1.0	2.4	280	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	31.7	23.0	30.5	23.2	29.3	22.9	25.5	29.4	24.8	28.7	24.0	27.9	3.5	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.2	28.4	23.4	18.2	27.9	22.6	17.2	27.2	78.3	29.5	75.3	28.9	72.0	28.0	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
10.4	9.1	7.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-14.1	35.4	4.7	2.0	-17.5	36.8	-20.3	38.0	-22.9	39.1	-26.3	40.5		
			-14.3	26.9	4.9	1.0	-17.8	27.6	-20.7	28.2	-23.5	28.8	-27.0	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	0.9	1.4	5.2	10.2	16.7	21.9	24.8	24.8	20.2	14.0	7.5	3.2
	DBStd	9.29	4.35	4.25	2.83	3.28	3.23	2.80	2.37	2.49	2.99	3.67	3.98	4.19
	HDD10.0	1019	282	240	151	35	1	0	0	0	0	11	90	210
	HDD18.3	2684	540	473	408	244	71	4	0	0	15	137	324	468
	CDD10.0	1983	0	0	2	42	208	358	457	458	307	136	16	1
	CDD18.3	608	0	0	0	1	19	113	199	199	72	4	0	0
	CDH23.3	4551	0	0	0	3	70	678	1727	1763	305	6	0	0
	CDH26.7	1193	0	0	0	1	8	140	482	524	39	0	0	0
Wind	WSAvg	3.2	3.1	3.4	3.6	3.4	3.1	2.9	2.8	2.8	3.3	3.2	3.3	3.4
Precipitation	PrecAvg	336	24	28	23	16	26	25	28	24	42	39	39	30
	PrecMax	663	70	67	64	59	78	77	120	78	123	243	117	74
	PrecMin	211	4	4	1	0	1	0	1	1	3	0	2	1
	PrecStd	95	18	18	15	12	17	17	26	18	31	46	30	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	10.4	16.1	22.6	27.9	31.9	33.8	34.2	29.7	23.9	17.6	12.5
		MCWB	6.4	6.7	10.0	14.8	18.8	21.9	22.4	22.3	22.6	18.2	14.1	9.9
	2%	DB	7.6	8.4	12.4	19.3	25.3	29.9	31.7	31.9	27.7	22.0	15.6	10.7
		MCWB	5.5	5.7	8.6	13.7	18.3	21.6	22.9	23.4	21.7	18.1	12.9	8.7
	5%	DB	6.6	7.1	10.4	17.1	23.4	28.3	30.4	30.7	26.2	20.7	14.2	9.6
		MCWB	5.0	5.2	7.5	12.9	17.9	21.2	23.1	23.7	21.1	17.4	12.3	8.1
	10%	DB	5.7	6.0	9.0	15.3	21.8	26.7	29.2	29.5	25.0	19.5	13.0	8.5
		MCWB	4.5	4.6	6.6	12.0	17.2	20.7	23.0	23.5	20.5	16.6	11.1	7.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	7.5	10.7	15.8	20.7	24.9	26.2	26.7	25.0	20.0	14.9	10.6
		MCDB	8.3	9.6	15.2	20.3	24.9	29.0	30.3	30.4	28.0	22.1	16.6	11.7
	2%	WB	6.0	6.2	9.2	14.4	19.3	23.3	25.3	25.7	23.1	18.9	13.4	9.1
		MCDB	7.1	7.6	11.9	18.2	23.5	27.7	29.3	29.5	26.2	21.0	15.0	10.3
	5%	WB	5.3	5.4	7.9	13.2	18.5	22.3	24.6	25.0	22.0	17.9	12.4	8.2
		MCDB	6.3	6.7	10.0	16.5	22.4	26.5	28.6	29.0	25.4	20.2	14.0	9.4
	10%	WB	4.6	4.8	7.0	12.2	17.6	21.4	23.8	24.3	21.0	16.9	11.3	7.1
		MCDB	5.5	5.9	8.6	14.9	21.2	25.6	27.9	28.2	24.0	19.2	12.9	8.3
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.2	5.4	6.2	6.9	7.5	7.6	7.6	7.0	6.7	6.0	5.3
		MCDBR	6.4	6.5	8.6	9.6	9.8	9.5	9.5	9.5	8.2	8.3	7.7	6.1
		MCWBR	4.9	4.7	5.2	5.0	4.7	4.7	4.8	5.2	4.5	5.3	5.7	4.6
	5% WB	MCDBR	5.6	5.6	7.8	8.9	8.4	8.1	7.9	8.4	7.5	7.3	6.8	4.9
		MCWBR	4.5	4.3	5.0	4.9	4.6	5.0	5.3	5.5	4.9	5.1	5.3	4.0
Clear-Sky Solar Irradiance	taub	0.368	0.405	0.445	0.509	0.480	0.480	0.512	0.513	0.477	0.423	0.383	0.374	
	taud	2.232	2.126	2.064	1.945	2.094	2.153	2.081	2.071	2.127	2.225	2.261	2.221	
	Ebn at Noon	734	765	784	764	802	802	770	753	747	740	712	684	
	Edh at Noon	93	122	146	178	158	149	159	155	137	110	90	87	
All-Sky Solar Radiation	RadAvg	1.42	2.19	3.23	4.53	5.53	6.16	6.06	5.42	4.16	2.65	1.71	1.26	
	RadStd	0.17	0.30	0.28	0.47	0.48	0.35	0.45	0.41	0.39	0.31	0.18	0.15	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page